Learning Deep Kernels for Non-Parametric Two-Sample Tests

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Abstract

We propose a class of kernel-based two-sample tests, which aim to determine whether two sets of samples are drawn from the same distribution. Our tests are constructed from kernels parameterized by deep neural nets, trained to maximize test power. These tests adapt to variations in distribution smoothness and shape over space, and are especially suited to high dimensions and complex data. By contrast, the simpler kernels used in prior kernel testing work are spatially homogeneous, and adaptive only in lengthscale. We explain how this scheme includes popular classifier-based two-sample tests as a special case, but improves on them in general. We provide the first proof of consistency for the proposed adaptation method, which applies both to kernels on deep features and to simpler radial basis kernels or multiple kernel learning. In experiments, we establish the superior performance of our deep kernels in hypothesis testing on benchmark and real-world data. The code of our deep-kernel-based two sample tests is available at github.com/fengliu90/DK-for-TST.

1. Introduction

Two sample tests are hypothesis tests aiming to determine whether two sets of samples are drawn from the same distribution. Traditional methods such as t-tests and Kolmogorov-Smirnov tests are mainstays of statistical applications, but require strong parametric assumptions about the distributions being studied and/or are only effective on data in extremely low-dimensional spaces. A broad set of recent work in statistics and machine learning has focused on relaxing these assumptions, with methods either generally applicable or specific to various more complex domains (Gretton et al., 2012a; Székely & Rizzo, 2013; Heller & Heller, 2016; Jitkrittum et al., 2016; Ramdas et al., 2017; Lopez-Paz & Oquab, 2017; Chen & Friedman, 2017; Gao et al., 2018; Ghoshdastidar et al., 2017; Ghoshdastidar & von Luxburg, 2018; Li & Wang, 2018; Kirchler et al., 2020). These tests have also allowed application in various machine learning problems such as domain adaptation, generative modeling, and causal discovery (Binkowski et al., 2018; Gong et al., 2016; Stojanov et al., 2019; Lopez-Paz & Oquab, 2017).

A popular class of non-parametric two-sample tests is based on kernel methods (Smola & Schölkopf, 2001): such tests construct a *kernel mean embedding* (Berlinet & Thomas-Agnan, 2004; Muandet et al., 2017) for each distribution, and measure the difference in these embeddings. For any *characteristic* kernel, two distributions are the same if and only if their mean embeddings are the same; the distance between mean embeddings is the *maximum mean discrepancy* (MMD) (Gretton et al., 2012a). There are also several closely related methods, including tests based on checking for differences in mean embeddings evaluated at specific locations (Chwialkowski et al., 2015; Jitkrittum et al., 2016) and kernel Fisher discriminant analysis (Harchaoui et al., 2007). These tests all work well for samples from simple distributions when using appropriate kernels.

Problems that we care about, however, often involve distributions with complex structure, where simple kernels will often map distinct distributions to nearby (and hence hard to distinguish) mean embeddings. Figure 1a shows an example of a multimodal dataset, where the overall modes align but the sub-mode structure varies differently at each mode. A translation-invariant Gaussian kernel only “looks at” the data uniformly within each mode (see Figure 1b), requiring many samples to correctly distinguish the two distributions. The distributions can be distinguished more effectively if we understand the structure of each mode, as with the more complex kernel illustrated in Figure 1c.

To model these complex functions, we adopt a *deep kernel* approach (Wilson et al., 2016; Sutherland et al., 2017; Li et al., 2017; Jean et al., 2018; Wenliang et al., 2019),

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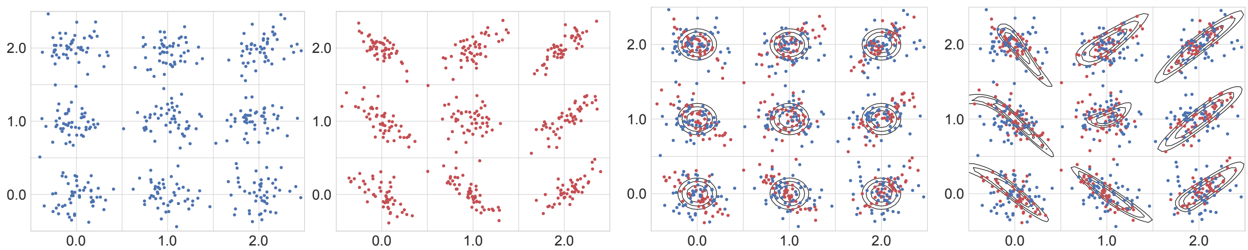

(a) Samples drawn from $\mathbb{P}$ (left) and $\mathbb{Q}$ (right). (b) Contour of Gaussian (c) Contour of deep kernel

Figure 1. In the Blob dataset, $\mathbb{P}$ and $\mathbb{Q}$ are each equal mixtures of nine Gaussians with the same modes (a), but each component of $\mathbb{P}$ is an isotropic Gaussian whereas the covariance of $\mathbb{Q}$ differs in each component. Panels (b) and (c) show the contours of a kernel, $k(x, \mu_i)$ for each of the nine modes μ_i; contour values are 0.7, 0.8 and 0.9. A Gaussian kernel (b) treats points isotropically throughout the space, based only on $\|x - y\|$. A deep kernel (c) learned by our methods behaves differently in different parts of the space, adapting to the local structure of the data distributions and hence allowing better identification of differences between $\mathbb{P}$ and $\mathbb{Q}$.

building a kernel with a deep network. In this paper, we use

$$k_\omega(x,y) = [(1-\epsilon)\kappa(\phi_\omega(x),\phi_\omega(y)) + \epsilon]q(x,y), \quad (1)$$

where the deep neural network ϕ_ω extracts features of samples, and κ is a simple kernel (e.g., a Gaussian) on those features, while q is a simple characteristic kernel (e.g. Gaussian) on the input space. With an appropriate choice of ϕ_ω, this allows for extremely flexible kernels which can learn complex behavior very different in different parts of space. This choice is discussed further in Section 5.

These complex kernels, though, cannot feasibly be specified by hand or simple heuristics, as is typical practice in kernel methods. We select the parameters ω by maximizing the ratio of the MMD to its variance, which maximizes test power at large sample sizes. This procedure was proposed by Sutherland et al. (2017), but we establish for the first time that it gives consistent selection of the best kernel in the class, whether optimizing our deep kernels with hundreds of thousands of parameters or simply choosing lengthscales of a Gaussian as did Sutherland et al. Previously, there were no guarantees this procedure would yield a kernel which generalized at all from the training set to a test set.

Another way to compare distributions is to train a classifier between them, and evaluate its accuracy (Lopez-Paz & Oquab, 2017). We show, perhaps surprisingly, that our framework encompasses this approach, but deep kernels allow for more general model classes which can use the data more efficiently. We also train representations directly to maximize test power, rather than a cross-entropy surrogate.

We test our method on several simulated and real-world datasets, including complex synthetic distributions, high-energy physics data, and challenging image problems. We find convincingly that learned deep kernels outperform simple shallow methods, and learning by maximizing test power outperforms learning through a cross-entropy surrogate loss.

2. MMD Two-Sample Tests

Two-sample testing. Let $\mathcal{X}$ be a separable metric space – in this paper, typically a subset of $\mathbb{R}^d$ – and $\mathbb{P}$, $\mathbb{Q}$ be Borel probability measures on $\mathcal{X}$. We observe independent identically distributed (*i.i.d.*) samples $S_\mathbb{P} = \{x_i\}_{i=1}^n \sim \mathbb{P}^n$ and $S_\mathbb{Q} = \{y_j\}_{j=1}^m \sim \mathbb{Q}^m$. We wish to know whether $S_\mathbb{P}$ and $S_\mathbb{Q}$ come from the same distribution: does $\mathbb{P} = \mathbb{Q}$?

We use the null hypothesis testing framework, where the null hypothesis $\mathfrak{H}_0 : \mathbb{P} = \mathbb{Q}$ is tested against the alternative hypothesis $\mathfrak{H}_1 : \mathbb{P} \neq \mathbb{Q}$. We perform a two-sample test in four steps: select a significance level $\alpha \in [0,1]$; compute a test statistic $\hat{t}(S_\mathbb{P}, S_\mathbb{Q})$; compute the p-value $\hat{p} = \Pr_{\mathfrak{H}_0}(T > \hat{t})$, the probability of the two-sample test returning a statistic as large as $\hat{t}$ when $\mathfrak{H}_0$ is true; finally, reject $\mathfrak{H}_0$ if $\hat{p} < \alpha$.

Maximum mean discrepancy (MMD). We will base our two-sample test statistic on an estimate of a distance between distributions. Our metric, the MMD, is defined in terms of a kernel k giving point-level "similarities" on $\mathcal{X}$.

Definition 1 (Gretton et al., 2012a)**.** *Let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be the kernel of a reproducing kernel Hilbert space $\mathcal{H}_k$, with feature maps $k(\cdot, x) \in \mathcal{H}_k$. Let $X, X' \sim \mathbb{P}$ and $Y, Y' \sim \mathbb{Q}$, and define the* kernel mean embeddings $\mu_\mathbb{P} := \mathbb{E}[k(\cdot, X)]$ *and $\mu_\mathbb{Q} := \mathbb{E}[k(\cdot, Y)]$. Under mild integrability conditions,*

$$\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) := \sup_{f \in \mathcal{H}, \|f\|_{\mathcal{H}_k} \leq 1} |\mathbb{E}[f(X)] - \mathbb{E}[f(Y)]|$$

$$= \|\mu_\mathbb{P} - \mu_\mathbb{Q}\|_{\mathcal{H}_k} = \sqrt{\mathbb{E}\left[k(X,X') + k(Y,Y') - 2k(X,Y)\right]}.$$

For characteristic *kernels, $\mu_\mathbb{P} = \mu_\mathbb{Q}$ implies $\mathbb{P} = \mathbb{Q}$, hence $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}; \mathcal{H}_k) = 0$ if and only if $\mathbb{P} = \mathbb{Q}$.*

The first form shows that the MMD is an integral probability metric (Müller, 1997), along with such popular distances as the Wasserstein and total variation.

There are several natural estimators of the MMD from samples. We will assume $n = m$ and use the U-statistic estimator, which is unbiased for MMD^2 and has nearly minimal variance among unbiased estimators (Gretton et al., 2012a):

$$\widehat{\mathrm{MMD}}_u^2(S_\mathbb{P}, S_\mathbb{Q}; k) := \frac{1}{n(n-1)} \sum_{i \neq j} H_{ij} \tag{2}$$

$$H_{ij} := k(X_i, X_j) + k(Y_i, Y_j) - k(X_i, Y_j) - k(Y_i, X_j).$$

The similar $\widehat{\mathrm{MMD}}_b^2 := \frac{1}{n^2} \sum_{ij} H_{ij}$ is the squared MMD between the empirical distributions of $S_\mathbb{P}$ and $S_\mathbb{Q}$.[1]

Testing with the MMD. It can be shown that under $\mathfrak{H}_0$, $n\widehat{\mathrm{MMD}}_u^2$ converges to a distribution depending on $\mathbb{P}$ and k; we thus use this as our test statistic.

Proposition 2 (Asymptotics of $\widehat{\mathrm{MMD}}_u^2$). *Under the null hypothesis, $\mathfrak{H}_0 : \mathbb{P} = \mathbb{Q}$, we have if $Z_i \sim \mathcal{N}(0, 2)$,*

$$n\widehat{\mathrm{MMD}}_u^2 \xrightarrow{d} \sum_i \sigma_i (Z_i^2 - 2);$$

here σ_i are the eigenvalues of the $\mathbb{P}$-covariance operator of the centered kernel (Gretton et al., 2012a, Theorem 12), and $\xrightarrow{d}$ denotes convergence in distribution.

Under the alternative, $\mathfrak{H}_1 : \mathbb{P} \neq \mathbb{Q}$, a standard central limit theorem holds (Serfling, 1980, Section 5.5.1):

$$\sqrt{n}(\widehat{\mathrm{MMD}}_u^2 - \mathrm{MMD}^2) \xrightarrow{d} \mathcal{N}(0, \sigma_{\mathfrak{H}_1}^2)$$
$$\sigma_{\mathfrak{H}_1}^2 := 4\left(\mathbb{E}[H_{12}H_{13}] - \mathbb{E}[H_{12}]^2\right)$$

where H_{12}, H_{13} refer to H_{ij} above.

Although it is possible to construct a test based on directly estimating this null distribution (Gretton et al., 2009), it is both simpler and, if implemented carefully, faster (Sutherland et al., 2017) to instead use a permutation test. This general method (Dwass, 1957; Alba Fernández et al., 2008) observes that under $\mathfrak{H}_0$, the samples from $\mathbb{P}$ and $\mathbb{Q}$ are interchangeable; we can therefore estimate the null distribution of our test statistic by repeatedly re-computing it with the samples randomly re-assigned to $S_\mathbb{P}$ or $S_\mathbb{Q}$.

Test power. The main measure of efficacy of a null hypothesis test is its *power*: the probability that, for a particular $\mathbb{P} \neq \mathbb{Q}$ and n, we correctly reject $\mathfrak{H}_0$. Proposition 2 implies, where Φ is the standard normal CDF, that

$$\Pr_{\mathfrak{H}_1}\left(n\widehat{\mathrm{MMD}}_u^2 > r\right) \to \Phi\left(\frac{\sqrt{n}\,\mathrm{MMD}^2}{\sigma_{\mathfrak{H}_1}} - \frac{r}{\sqrt{n}\,\sigma_{\mathfrak{H}_1}}\right);$$

we can find the approximate test power by using the rejection threshold, found via (e.g.) permutation testing, as r. We also know via Proposition 2 that this r will converge to a constant, and MMD, $\sigma_{\mathfrak{H}_1}$ are also constants. For reasonably large n, the power is dominated by the first term, and the kernel yielding the most powerful test will approximately maximize (Sutherland et al., 2017)

$$J(\mathbb{P}, \mathbb{Q}; k) := \mathrm{MMD}^2(\mathbb{P}, \mathbb{Q}; k) / \sigma_{\mathfrak{H}_1}(\mathbb{P}, \mathbb{Q}; k). \tag{3}$$

Selecting a kernel. The criterion $J(\mathbb{P}, \mathbb{Q}; k)$ depends on the particular $\mathbb{P}$ and $\mathbb{Q}$ at hand, and thus we typically will neither be able to choose a kernel *a priori*, nor exactly evaluate J given samples. We can, however, estimate it with

$$\hat{J}_\lambda(S_\mathbb{P}, S_\mathbb{Q}; k) := \frac{\widehat{\mathrm{MMD}}_u^2(S_\mathbb{P}, S_\mathbb{Q}; k)}{\hat{\sigma}_{\mathfrak{H}_1,\lambda}(S_\mathbb{P}, S_\mathbb{Q}; k)}, \tag{4}$$

where $\hat{\sigma}_{\mathfrak{H}_1,\lambda}^2$ is a regularized estimator of $\sigma_{\mathfrak{H}_1}^2$ given by[2]

$$\frac{4}{n^3} \sum_{i=1}^n \left(\sum_{j=1}^n H_{ij}\right)^2 - \frac{4}{n^4}\left(\sum_{i=1}^n \sum_{j=1}^n H_{ij}\right)^2 + \lambda. \tag{5}$$

Given $S_\mathbb{P}$ and $S_\mathbb{Q}$, we could construct a test by choosing k to maximize $\hat{J}_\lambda(S_\mathbb{P}, S_\mathbb{Q}; k)$, then using a test statistic based on $\widehat{\mathrm{MMD}}(S_\mathbb{P}, S_\mathbb{Q}; k)$. This sample re-use, however, violates the conditions of Proposition 2, and permutation testing would require repeatedly re-training k with permuted labels.

Thus we split the data, get $k^{tr} \approx \arg\max_k \hat{J}_\lambda(S_\mathbb{P}^{tr}, S_\mathbb{Q}^{tr}; k)$, then compute the test statistic and permutation threshold on $S_\mathbb{P}^{te}$, $S_\mathbb{Q}^{te}$ using k^{tr}. This procedure was proposed for $\widehat{\mathrm{MMD}}_u^2$ by Sutherland et al. (2017), but the same technique works for a variety of tests (Gretton et al., 2012b; Jitkrittum et al., 2016; 2017; Lopez-Paz & Oquab, 2017). Our paper adopts this framework (Section 5) and studies it further.

Relationship to other approaches. One common scheme is to pick a kernel k_ω based on some proxy task, such as a related classification problem (e.g. Kirchler et al. 2020 or the KID score of Binkowski et al. 2018). Although this approach can work quite well, it depends entirely on features from the proxy task applying well to the differences between $\mathbb{P}$ and $\mathbb{Q}$, which can be hard to know in general.

An alternative is to maximize simply $\widehat{\mathrm{MMD}}_u$ (Sriperumbudur et al. 2009; proposed but not evaluated by Kirchler

[1] Including $k(X_i, Y_i)$ terms in $\widehat{\mathrm{MMD}}_u$ gives the minimal variance unbiased estimator, and allows $m \neq n$. The U-statistic is more convenient for analysis and for efficient permutations; in our settings it behaves similarly to the MVUE and $\widehat{\mathrm{MMD}}_b^2$.

[2] This estimator, as a V-statistic, is biased even when $\lambda = 0$ (although this bias is only $O(1/N)$; see Lemma 18). Although Sutherland et al. (2017); Sutherland (2019) give a quadratic-time estimator unbiased for $\sigma_{\mathfrak{H}_1}^2$, it is much more complicated to implement and analyze, likely has higher variance, and (being unbiased) can be negative, especially e.g. when the kernel is poor.